Directing the Room's AI Use: An SRL Guide

HONR 46400 · Evidence-Driven Research

How to put AI to work in a session you are leading, without letting it do the thinking for the room. This is the practical companion to the handbook's philosophy. Read it before your first Monday slot, where AI is the center of the investigation.

The one rule underneath all the others

You are teaching the room to use AI the way this course wants: as an arm, not a brain. That means every time you point the class at an AI tool, you are staging a small demonstration of good research behavior. The class commits first, asks the AI something specific, then checks what it gets back against real evidence, and only then decides. Your job is to run that loop out loud so everyone sees the seams.

If you take nothing else from this guide: **the AI's answer is never the last word.** A human decision is.

Same prompt or split roles?

You have two ways to put the room on a prompt. Choose on purpose.

Have everyone run the same prompt when you want to expose variation. Five people typing the identical prompt into the same AI tool will often get five slightly different answers, and sometimes one wildly different one. That spread is the lesson: it shows the room that an AI answer is not a fixed fact you looked up, it is a generated response that could have come out otherwise. Use this when the point is "do not trust a single run."

Split the room into roles when you want depth on a claim. Assign jobs and have people bring their piece back:

- One person asks the AI to **make the case** for a claim.
- Another asks it to **argue the opposite**.
- Another asks it to **find the sources** and then goes to confirm they exist.
- Another asks it to **name what it is uncertain about**.

Splitting roles turns a single answer into an argument the room can referee. Use it on Wednesday's applied lab, where you have a longer block and want the class working hands-on rather than watching one screen.

A quick rule of thumb: **same prompt to show that AI is not authoritative; split roles to interrogate a claim from every side.**

Running a live human-versus-AI comparison

This is the signature Monday move, and it only works if the order is right.

1. **Commit first.** The room writes its answer before the AI is opened. If you skip this, the comparison is worthless, because the AI's answer will quietly become everyone's answer.
2. **Ask the AI the same question.** Use the specific prompt from your prep template, projected where the room can see it.
3. **Put them side by side.** The room's committed answer on one side, the AI's on the other. Read both out loud.
4. **Work the gap.** Where they agree, ask *why*: agreement can mean you are both right or both wrong. Where they differ, ask *who is right and how we would know*. Drive the room toward evidence, not toward the more confident voice.
5. **Decide as humans.** The room commits to a final answer that it can defend. The AI does not get a vote; it got consulted.

The magic is in step 4. Do not resolve the disagreement for the room. Make them resolve it with evidence.

The three AI-failure patterns to hunt for

Every session, you are hunting for three specific ways AI goes wrong. Name them for the room; teach the class to spot them.

1. Confident fabrication

The AI invents something that does not exist and states it with total confidence: a study, an author, a statistic, a quotation, a dataset. This is the most dangerous pattern because the confidence is identical to when it is right.

How to hunt it: whenever the AI hands you a source or a specific number, stop and confirm it independently. "It cited a 2019 paper. Someone find it. Does it exist? Does it say this?" If the class cannot retrieve it, the AI may have made it up. In this course, an unverifiable source is treated as a fabrication, full stop.

2. Plausible-but-wrong method

The AI proposes an approach that sounds reasonable and is subtly wrong for the job: the wrong comparison, a design that cannot answer the question asked, a metric that measures the wrong thing, an analysis that leaks future information.

How to hunt it: ask the room whether the method actually answers the question the room asked. "It gave us a way to do this. Does this design let us claim what we want to claim, or something weaker?" The tell is that the method is real, just mismatched to the goal.

3. Missing uncertainty

The AI states a finding with no hedge, no limitation, and no sense of how sure it should be. It reports a point where it should report a range, or a conclusion where it should report a caveat.

How to hunt it: ask "where does this answer tell us what it does not know?" If the answer is nowhere, that silence is the defect. Real findings come with uncertainty attached. An answer with none is overclaiming, and overclaiming is a failure archetype this course grades against.

Using the "After running, verify" checklist as a class activity

Every AI prompt in the course notebooks is followed by an **"After running, verify"** checklist. Do not let the class skip past it. Turn it into a live, 30-second activity the room does together, out loud, after every prompt you run.

Run it as three quick calls to the room:

- **Confirm the sources exist.** "It named a source. Is it real? Confirm it before we use it."
- **Cross-check the facts.** "Does any claim here contradict what we already verified? Check it against something we trust."
- **Log the decision.** "What did we decide because of this, and how did we check it? That is our Ledger line."

That last step feeds the **AI Research Ledger**, the record every session keeps of what AI did and how it was checked. Say the Ledger line out loud and have someone write it down. A session that runs prompts but never verifies them has modeled exactly the behavior the course is trying to break.

When the AI is right and the class is wrong

This will happen, and it is one of the best teaching moments you will get. The room commits to an answer, the AI disagrees, and the AI turns out to be correct.

Do not gloat, and do not let the room feel ambushed. Turn it into a lesson about *how you knew*:

- "The AI disagreed with us. Before we assume it is right, how would we check?"
- Make the room find the evidence that confirms the AI, not just accept the AI's word. The point is not that the AI won. The point is that **evidence** settled it, and this time the evidence happened to be on the AI's side.
- Then close the loop: "So we were wrong. What assumption led us there?" A wrong prior that you examined is worth more than a right one you never questioned.

This is the mirror image of hunting AI failures. The AI is a tool that is sometimes right and sometimes wrong, and the researcher's job is the same either way: verify against evidence, then decide.

Never let the AI end the discussion

Watch for the moment the room goes quiet because "the AI already answered it." That silence is the failure this whole system exists to prevent. When you feel it, break it:

- “The AI gave us an answer. We have not decided anything yet. What do *we* think?”
- “Good, that is the AI’s version. Now defend it or beat it.”
- “We are not done because a tool produced text. We are done when a person commits and holds up under a hard question.”

Close every AI block on a human decision, recorded in the Ledger and the Claim Ticket. The AI proposes. The room verifies. A person decides and defends. That is the course in miniature, and every session you lead is one more rehearsal of it.

Related files: `srl_handbook.md` *(the full role),* `socratic_question_bank.md` *(the “compare human vs AI” and “expose AI overconfidence” sets map directly onto this guide), and* `srl_rubric.md` *(rows 4 and 5 grade exactly what this guide teaches).*